

# K.M.S.A. DARSHANA

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## Professional Summary

Information Technology undergraduate at SLIIT with a strong drive toward **software development**, having independently and collaboratively delivered **7+ full-stack and mobile applications** using modern technologies including React.js, Node.js, Spring Boot, and Kotlin. Specialized in Information Technology with hands-on focus on real-world software engineering — building scalable web systems, Android apps, and NLP-integrated platforms. Currently seeking a **software development internship** to apply project experience, grow within a professional engineering team, and contribute meaningfully from day one.

## Education

**Sri Lanka Institute of Information Technology (SLIIT)** | Malabe, Sri Lanka 2023 – Present  
Bachelor of Science Honours in Information Technology — Specialization: Information Technology

- Current CGPA: **3.22 / 4.00**
- Relevant Coursework: Data Structures & Algorithms, OOP, Database Management, Mobile App Development, Web Technologies, Software Architecture

## Experience

**Group Project Contributor & Full-Stack Developer** | SLIIT — Academic Team Projects 2023 – Present  
Collaborative Software Engineering Experience

- Contributed to **4+ multi-member team projects** at SLIIT, collaborating with cross-functional groups of 3–5 members to design, build, and deliver full-stack software systems on deadline.
- Applied Agile-style workflows across team projects — managing task allocation, conducting code reviews, and resolving merge conflicts using **Git and GitHub** in collaborative repositories.
- Took ownership of backend architecture in team settings, implementing **REST APIs, JWT authentication, and MongoDB schemas** that other team members integrated with their frontend modules.
- Coordinated UI/UX responsibilities in group projects using **Figma** for wireframing before development, reducing miscommunication and rework between team members.
- Communicated technical decisions and progress updates within teams, strengthening both written and verbal collaboration skills in a software engineering context.

## Technical Skills

|                        |                                                                              |
|------------------------|------------------------------------------------------------------------------|
| Languages              | JavaScript, Java, Kotlin, PHP, SQL, HTML5, CSS3, XML                         |
| Frameworks & Libraries | React.js, Next.js, Node.js, Express.js, Spring Boot, Tailwind CSS, Bootstrap |
| Mobile                 | Android Studio, Kotlin, SharedPreferences                                    |
| Databases              | MongoDB, MySQL, Oracle, Firebase                                             |
| Tools & Platforms      | Git, GitHub, VS Code, Figma, Vercel, Tomcat, Maven, Postman, EmailJS         |
| AI / ML                | Sentence Transformers (SBERT), all-MiniLM-L6-v2, Cosine Similarity, NLP      |
| Concepts               | REST API, JWT Authentication, OAuth2, MVC, RBAC, OOP, Agile                  |

## Projects

**Smart Campus Operations Hub** May 2026 | GitHub  
React 19, Spring Boot 3, MongoDB, OAuth2, JWT, Tailwind CSS, Maven, Axios

- Architected a full-stack campus management platform supporting **3 distinct user roles** (User/Technician/Admin), delivering end-to-end resource discovery, booking, and SLA tracking across a single unified interface.
- Implemented a QR-based check-in system with real-time notifications, improving operational efficiency across campus facilities.
- Fortified system security by integrating OAuth2 and JWT authentication alongside login audit logging, securing **all platform entry points** against unauthorized access.

Lost & Found System

Apr 2026 | GitHub | Live Demo

Next.js, React, MongoDB, Mongoose, Tailwind CSS, Node.js, JWT, SBERT, all-MiniLM-L6-v2

- Designed and launched a Smart Campus Lost & Found platform handling structured item reporting, claim management, and admin monitoring across a **5-step verification workflow**.
- Integrated **Sentence Transformers (SBERT)** with the all-MiniLM-L6-v2 pretrained model to generate semantic embeddings of item descriptions, using **cosine similarity** to automatically match lost items with claims and flag low-confidence or fraudulent submissions.
- Delivered interactive data visualizations for administrators, enabling faster decision-making across **multiple item categories and claim statuses**.

Hospital Management System

Aug 2025 | GitHub

React 18, Node.js, Express.js, MongoDB, JWT, Bcrypt, Tailwind CSS, Vite, PDFKit

- Engineered a comprehensive MERN-stack Hospital Management System covering **5 core modules**: patient management, appointment scheduling, lab test tracking, pharmacy, and staff management.
- Strengthened data protection by implementing RBAC across **4+ staff roles**, securing sensitive medical records with JWT and Bcrypt encryption.
- Accelerated reporting workflows by integrating PDFKit-powered PDF export, enabling **one-click generation** of patient records and operational summaries without manual formatting.

Online Customer Care System (CareDesk)

Apr 2025 | GitHub

Java Servlets, JSP, JDBC, MySQL, Apache Tomcat, MVC

- Constructed an enterprise-grade ticketing platform with Java Servlets and JSP, enabling customers to submit, track, and rate support tickets through a **3-stage resolution pipeline**.
- Structured the system using an MVC architecture, allowing admins to manage **multiple ticket queues** and monitor customer satisfaction metrics in real time.

Blood Donation Management System

Jul 2024 | GitHub

PHP, MySQL, JavaScript, jQuery, HTML5, CSS3

- Launched a full-stack blood donation platform connecting **3 user types** (donors, recipients, admins) with secure authentication and real-time blood inventory management.
- Streamlined donation processing by automating appointment scheduling and admin approval workflows, significantly reducing manual coordination effort across the platform.

Android Mobile Apps — Daily Habit Tracker & Paw Finder

Aug 2025 | Tracker | Paw Finder

Kotlin, XML, Android Studio, SharedPreferences

- **Daily Habit Tracker**: Crafted an Android wellness app tracking **4 health dimensions** (habits, mood, hydration, analytics) with daily reminders, boosting user consistency through streak-based engagement.
- **Paw Finder**: Deployed a feature-rich Android pet-community app connecting lost pet owners with finders and adopters, incorporating **3 core modules**: lost pets, adoptions, and community feed.

Languages

Sinhala — Native | English — Professional Proficiency